



Mahindra University Hyderabad
École Centrale School of Engineering
Minor-I exam

Program: B. Tech. Branch: CSE, AI, ICS, CAM, ECE, VLSI, COM Year: II Semester: II
Subject: Theory of Computation (CS2204)

Date: 24-02-2026
Time Duration: 1.5 Hours

Start Time: 2:00 PM
Max. Marks: 50

Instructions:

- 1) All parts of a question should be answered consecutively.
- 2) Any question attempted using pencil will not be considered for the evaluation.
- 3) Mobile phones and computers of any kind should not be brought inside the exam hall.
- 4) Use of any unfair means will result in severe disciplinary action.

Q1.

[2*3=6 Marks]

- 1.1. Given a relation $R = \{(1,2), (2,1), (2,3), (3,4), (4,2)\}$ on a set $A = \{1,2,3,4\}$, Determine whether R is (i) symmetric (ii) anti-symmetric. Show with detail supportive steps.
- 1.2. Consider the relation $R = \{(1,1), (2,2), (2,3), (3,1), (1,4), (2,4)\}$ on a set of $A = \{1,2,3,4\}$. Obtain R^* , the reflexive transitive closure of R .

Q2.

[2*6=12 Marks]

Construct a Deterministic Finite Automaton (DFA) for each of the following languages defined over the alphabet $\Sigma = \{a,b\}$. Also provide the formal 5-tuple definition of the DFA along with the transition table.

- 2.1. $L_1 = \{w \mid w \text{ ends with } aa\}$ with three states.
- 2.2. $L_2 = \{w \mid w \text{ does not end with } aa\}$ with three states

Q3. Consider the given DFA transition table and draw the corresponding minimized DFA along with the transition table.

[2*6=12 Marks]

- 3.1. States: $\{A, B, C, D, E, F, G\}$; Alphabet: $\{0,1\}$; Start: A ; Final States: $\{D, F\}$

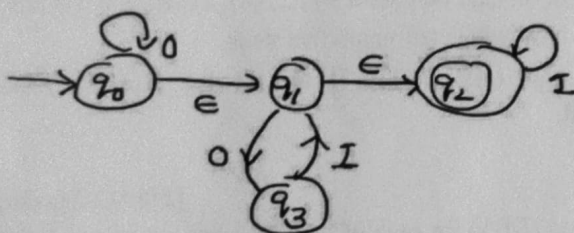
State	0	1
A	B	C
B	A	D
C	E	F
D	D	D
E	G	F
F	F	F
G	G	F

3.2. States: $\{S1, S2, S3, S4, S5, S6, S7, S8\}$; Alphabet: $\{0, 1\}$; Start: $S1$; Final States: $\{S6, S7\}$

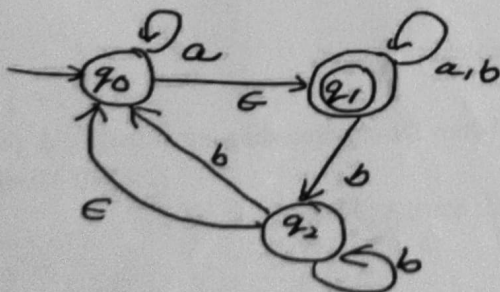
State	0	1
S1	S2	S3
S2	S4	S5
S3	S4	S5
S4	S6	S7
S5	S6	S7
S6	S6	S8
S7	S6	S8
S8	S8	S8

Q4. Consider the given non-deterministic finite automata (NFA) with ϵ -moves convert it into its equivalent NFA and then convert the constructed NFA into DFA. [2*7=14 Marks]

4.1.



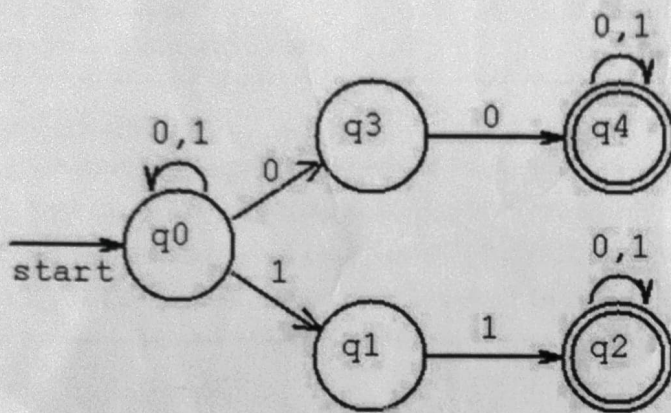
4.2.



Q5.

Convert the following NFA to DFA. Also ensure that no unreachable states are present in the resultant DFA.

[1*6=6 Marks]



***** All The Best *****